

Hypercalcaemia in adults

NICE NG132, 2019, DTB 2010;48:30, JAMA 2022;328(16):1624, BMJ 2015;350:h2723
Derbyshire Shared Care Pathology Guidelines, (hypercalcaemia and primary hyperparathyroidism) accessed June 2023



Red Whale

GEMS
Guidelines & Evidence Made Simple

In this GEMS, all 'calcium' values refer to 'corrected calcium' (adjusted for albumin)

When to check calcium?

Check corrected calcium if ANY of:

- Symptoms of hypercalcaemia: thirst, frequency of/excessive urination, constipation, nausea, vomiting, drowsiness (late sign).
- Any other features that may indicate primary hyperparathyroidism: bone pathology (e.g. osteoporosis, previous fragility fracture) or renal pathology (e.g. a renal stone).
- A known or suspected cancer that is associated with hypercalcaemia, e.g. lung cancer, myeloma.

Consider checking corrected calcium if ANY chronic non-differentiated symptoms:

- Bone, muscle or joint pain.
- Anxiety, depression, low mood, irritability, apathy or insomnia.
- Digestive problems, fatigue or mild confusion.

What to do with a calcium result?

Calcium 2.50–2.59mmol/l
(in the NORMAL range)

Calcium 2.60–2.99mmol/l

Calcium ≥ 3.00 mmol/l

90% of hypercalcaemia due to primary hyperparathyroidism or cancer

AND any features of primary hyperparathyroidism as above

Any symptoms of hypercalcaemia (above)?

Yes

Discuss with secondary care that day: may need admission for rehydration/treatment or may need urgent referral

No

Repeat corrected calcium. If on 2 separate occasions:

Corrected calcium ≥ 2.6 mmol/l

OR

Corrected calcium 2.50–2.59mmol/l AND any features of primary hyperparathyroidism as above

Arrange PTH and repeat corrected calcium again... What about other tests?

Vitamin D: NICE says do not check vitamin D or do any other tests when assessing for primary hyperparathyroidism. Local guidance often suggests checking vitamin D to assess for the underlying cause of hypercalcaemia.

Other useful tests to assess for underlying cause: rest of bone profile and U&Es. TFTs and myeloma screen may be appropriate. To assess for familial causes (see below): prolactin if <45y, calcium excretion index (urine sample).

PTH low
(below lower limit of reference range)

Non PTH-dependent hypercalcaemia

Causes include:

- Cancer (e.g. lung, myeloma, bony mets)
- Hyperthyroidism
- Vitamin D toxicity
- Sarcoid
- Thiazides
- Acromegaly
- Pheochromocytoma

PTH below midpoint of reference range (but within reference range) AND corrected calcium <2.6mmol/l

Primary hyperparathyroidism unlikely

PTH below midpoint of reference range AND corrected calcium ≥ 2.6 mmol/l

Cannot rule out primary hyperparathyroidism – seek endocrine advice

PTH raised or above midpoint of reference range AND primary hyperparathyroidism suspected

Likely primary hyperparathyroidism – seek endocrine advice (referral/telephone call)

Other causes include tertiary hyperparathyroidism (seen in renal disease), familial conditions, e.g. MEN 1/familial hypocalciuric hypercalcaemia, and lithium treatment

If on thiazide diuretic: stop. If excess dietary calcium: reduce. If vitamin D low: do not delay referral

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